

Last Time:

• V v.s.

(1) $v_1, \dots, v_n \in V$ is l.d. if $\exists (a_1, \dots, a_n) \in \mathbb{R}^n \setminus \{0\}$

s.t. $\sum_{i=1}^n a_i v_i = 0.$

(1') $v_1, \dots, v_n \in V$ is l.i. if not l.d.

(i.e., $\sum_{i=1}^n a_i v_i = 0 \Rightarrow a_i = 0 \ \forall i$)

(2) Subset $S \subseteq V$ is l.d. if

$\exists T = \{v_1, \dots, v_k\} \subseteq S$ s.t. $T \neq \emptyset$ (so $k \geq 1$)

and $v_1, \dots, v_k$ is l.d. as in (1).

(2') $S \subseteq V$ is l.i. if not l.d. ($\emptyset$ is l.i.)

(means every fin subset of S is l.i.)

Today:

(1) Finish §1.3 Linear (In)dependence (Properties)

(2) Start §1.4 Basis and Dimension

Prop 1.39: V v.s., $S_1 \subseteq S_2 \subseteq V$. S_1 l.d. $\Rightarrow S_2$ l.d.

Pf: S_1 l.d. $\Rightarrow \exists T = \{v_1, \dots, v_k\} \subseteq S_1$ s.t. $v_1, \dots, v_k$ l.d.

Since $T \subseteq S_1 \subseteq S_2 \xRightarrow{T \subseteq S_2} S_2$ l.d. via T by def. $\square$

$\Downarrow$ Contrapositive

Cor 1.40: V v.s., $S_1 \subseteq S_2 \subseteq V$. S_2 l.i. $\Rightarrow S_1$ l.i.

Let's compare "redundancy" with l.i.

$$\text{(e.g. } \text{span}(\overset{\text{l.i.}}{[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}]}], \overset{\text{l.i.}}{[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}]}]) = \text{span}(\overset{\text{l.d.}}{[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}]}], \overset{\text{l.d.}}{[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}]}], [\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}]}])$$

Lem 1.41: V v.s., $S \subseteq V$, $v \in S$. $v \in \text{span}(S \setminus \{v\})$

$\Rightarrow S$ l.d. $\{w \in S : w \neq v\}$

Pf: $v \in \text{span}(\underbrace{S \setminus \{v\}}_{\text{}})$ then $\exists T = \{w_1, \dots, w_k\} \subseteq S \setminus \{v\}$

s.t. $v = \sum_{i=1}^k a_i w_i$ for some coeffs $a_1, \dots, a_k \in \mathbb{R}$.

Then $a_1 w_1 + \dots + a_k w_k + \overset{\neq 0}{(-1)} v = 0 \Rightarrow w_1, \dots, w_k, v$ l.d.

$\Rightarrow S$ l.d. b/c $\{w_1, \dots, w_k, v\} \subseteq S$. $\square$

Prop 1.42: V v.s., $S \subseteq V$, $v \in S$. Then

$\text{span}(S \setminus \{v\}) = \text{span}(S) \overset{\text{iff}}{\iff} v \in \text{span}(S \setminus \{v\})$.

Pf: $(\Rightarrow)$: $\text{span}(S \setminus \{v\}) = \text{span}(S) \supseteq S \ni v$. $\square$

$(\Leftarrow)$: $v \in S$ and $v \in \text{span}(S \setminus \{v\})$

$\Rightarrow S = (S \setminus \{v\}) \cup \{v\} \subseteq \text{span}(S \setminus \{v\})$. So Prop 1.33

$\Rightarrow \text{span}(S) \subseteq \text{span}(S \setminus \{v\})$.

Also, $S \setminus \{v\} \subseteq S \subseteq \text{span}(S)$ by Prop 1.33

$\Rightarrow \underline{\text{span}(S \setminus \{v\}) \subseteq \text{span}(S)}$ by Prop 1.33.

Thus, $\text{span}(S \setminus \{v\}) = \text{span}(S)$. $\square$

Prop 1.43: V v.s., $S \subseteq V$. S l.d. iff $\exists v \in S$
s.t. $\text{span}(S \setminus \{v\}) = \text{span}(S)$.

Pf: ($\Leftarrow$): Lem 1.41. $\square$

($\Rightarrow$): Unit 1.3 Notes. $\square$

Rmk: Such $v \in S$ from Prop 1.43 is a
"redundant" vec of S to turn S into l.d. subset,
and removing v from S does not change span.

L.i. analog of Lem 1.41:

Lem 1.44: V v.s., $S \subseteq V$ l.i., $v \in V$. If $v \notin \text{span}(S)$,
then $S \cup \{v\}$ l.i.

Pf when $S = \{w_1, \dots, w_n\}$ fin:

$$v \notin \text{span}(S) \Rightarrow v \notin S = \{w_1, \dots, w_n\} \Rightarrow \underline{S \cup \{v\}} = \{w_1, \dots, w_n, v\}$$

Assume by contradiction that $\underline{w_1, \dots, w_n, v}$ is l.d.

Then $\exists (a_1, \dots, a_n, b) \in \mathbb{R}^{n+1} \setminus \{0\}$ s.t. $\sum_{i=1}^n a_i w_i + b v \stackrel{(*)}{=} 0$.

If $b=0$, then $(*)$: $\sum_{i=1}^n a_i w_i = 0$ & $\underbrace{(a_1, \dots, a_n, 0)}_{\in \mathbb{R}^n \setminus \{0\}} \in \mathbb{R}^{n+1} \setminus \{0\}$

$\Rightarrow S = \{w_1, \dots, w_n\}$ l.d., contradiction to S l.i.

Thus, $b \neq 0$, so $(*) \Rightarrow b v = - \sum_{i=1}^n a_i w_i = \sum_{i=1}^n -a_i w_i$

$\stackrel{b \neq 0}{\Rightarrow} v = \sum_{i=1}^n \left(-\frac{a_i}{b}\right) w_i \in \text{span}(S)$, ~~$\times$~~ to $v \notin \text{span}(S)$.

Henceforth, $S \cup \{v\} = \{w_1, \dots, w_n, v\}$ l.i. $\square$

§1.4 Basis and Dimension

Motivation: V v.s., $W \subseteq V$ subsp. Then

$$W = \text{span}(W) = \text{span}(\underline{W \setminus \{0\}}).$$

l.d. unless $W = \{0\}$

It would be nice if we can find $S \subseteq W$ l.i.

s.t. $\text{span}(S) = W$. Then $\forall v \in S$,

$\text{span}(S \setminus \{v\}) \subsetneq \text{span}(S) = W$ by Prop 1.43

$\Rightarrow \underline{S}$ is a ^{min'l} minimal subset of W that spans W .

$\uparrow$ l.i. & spanning set of W ($= V$ as special case)
both

Q: How to find such S ?

Def 1.45: V v.s.

(a) A basis $\beta = (v_1, \dots, v_n) \in V^n$ of V is a l.i. list that also spans V .

(b) A basis $\beta \subseteq V$ of V is a l.i. subset that also spans V .

(two defs agree for $(v_1, \dots, v_n)$ & $\{v_1, \dots, v_n\}$)
ordered unordered

Ex 1.46:

(1) $\emptyset$ is a basis of $\{0\}$.

(2) The std basis of $\mathbb{R}^n$ is $\beta := (e_1, \dots, e_n)$ where $e_1, \dots, e_n$ are from Ex 1.38(1).

Pf: β l.i. b/c of Ex 1.38(1). Also,

$$\begin{aligned} \text{span}(\beta) &= \left\{ a_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + a_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} : a_i \in \mathbb{R} \forall i \right\} \\ &= \left\{ \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} : a_i \in \mathbb{R} \forall i \right\} = \mathbb{R}^n \Rightarrow \beta \text{ spans } \mathbb{R}^n \end{aligned}$$

Therefore, β is a basis of $\mathbb{R}^n$. □

(3) The std basis of $\mathbb{R}[x]_{\leq n}$ is $\beta := (1, x, \dots, x^n)$.

Pf: Exercise. □

(4) $(1, x+1, (x+1)^2)$ is a basis of $\mathbb{R}[x]_{\leq 2}$.

Pf: Exercise

□

Comment: Basis is not unique in gen'l! In

Ex 1.46 (3)(4), both $(1, x, x^2)$ & $(1, (x+1), (x+1)^2)$

are bases of $\mathbb{R}[x]_{\leq 2}$

↑ plural of basis